ZS-A14

Z-Spin-Mediated Singlet-to-Triplet Conversion: SU(2)-Covariant Usadel Closure, the Envelope-Shifted Current-Phase Relation, Three-Observable Locking, and Two-Channel Z-Sector Length Spectroscopy

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Version: v2.0 (Current-Phase Correction & Value-Maximizing Release) | Supersedes: v1.2

Verification: 47/47 PASS | Zero Free Parameters | ΔBIC = 8.28 (supporting) | MC hit = 0.64% < 5%

§0. Abstract

Version 2.0 corrects the single substantive error of v1.2 and converts it into two new theorems, raising the verification suite to 47/47 PASS with no new free parameters beyond the geometric impedance **A** = 35/437 and register **Q** = 11. The error: with a finite diffusive decay envelope the full Josephson critical current does not peak at the conversion optimum $d_F^* = (\pi/\mathbf{A})\xi_Z$; the envelope shifts the peak earlier. **Theorem A14.8′** (Envelope-Shifted Peak, DERIVED) gives the exact condition tan(**A** $d_F$peak/2ξZ) = **A** ξtZ/ξZ, so that only the conversion prefactor and the detrended current Ic·exp(+dF/ξtZ) peak at dF*. **Theorem A14.9′** (Three-Observable Locking, DERIVED) is restated in envelope-immune form: η, ρs^eff and the detrended current share the peak dF*, while the nodes of the raw current coincide with the η-nodes at dF = n·2πξZ/**A**, giving a common period ΔdF = 2dF* independent of the decay envelope. **Theorem A14.11** (Two-Channel Z-Length Spectroscopy, DERIVED/TESTABLE) turns the peak shift into a second probe: a single thickness scan yields ξZ from the node spacing and the Z-aligned triplet length ξtZ from the peak shift, ξtZ = (ξZ/**A**) tan(**A** dFpeak/2ξZ). The SU(2)-covariant Usadel theorem A14.7 is honestly relabelled DERIVED-CONDITIONAL: the equation form is IMPORTED-PROVEN from Bergeret–Tokatly and Tokatly, while the Z-Spin background field $\mathscr{A}_i$Z = (**A**/2ξZ)n̂Z·σ is a zero-parameter ansatz whose physical reality is the testable content. Tokatly's PROVEN dephasing tensor Γ^ab still lifts the angular factor sin²β of A14.4′ to a derived projection identity. Reference [2] is corrected to Yates et al. (arXiv:1606.08619). The BIC analysis is demoted to supporting evidence; the paper's falsifiable value is carried by the thickness-scan and peak-shift predictions.

§0.1 Epistemic Status Legend

Table 0.1. Epistemic Status Legend (Z-Spin convention).

Status	Definition
PROVEN	Mathematical theorem; standard math alone, machine / 50-digit precision.
IMPORTED-PROVEN	Result proved in an external work and used here without re-proof (e.g. SU(2) Usadel).

Status	Definition
DERIVED	From PROVEN items + Z-Spin axioms; zero free parameters beyond A = 35/437.
DERIVED-CONDITIONAL	DERIVED modulo one explicit conditioning item (here: physical reality of $\mathscr{A}_i$Z, or the value of ξZ), with closure / falsification path.
DERIVED-under-Regge	Conditional on the Regge-lattice framework (inherited from ZS-A12).
VERIFIED	Numerical / empirical confirmation at the stated precision.
TESTABLE	Pre-registered quantitative prediction with explicit falsification condition.
LOCKED	Core constant fixed upstream; not adjustable here.
NON-CLAIM	Explicit statement of what this paper does NOT establish.
OPEN	Recognized gap with explicit closure path.

§1. Introduction and v2.0 Correction Summary

Version 1.2 closed four extension topics but asserted that the full critical current Ic(dF) peaks at the conversion optimum dF*. An external review correctly observed that, for the closed form of NEW-20, the finite diffusive envelope exp(−dF/ξtZ) shifts the peak to dF^peak < dF* (numerically ≈ 5.8 ξZ at β = 0 and ≈ 3.1 ξZ at β = π/4, versus dF* ≈ 39.23 ξZ). v2.0 corrects every statement that conflated the prefactor peak with the full-current peak, and promotes the correction to two theorems that make A14 a sharper experimental tool than before.

The unifying object is unchanged — a single dimensionless Z-cell phase φZ(dF) = **A** dF/ξZ controls every observable — but v2.0 distinguishes three layers of that control: (i) the *prefactor* $\sin^2$(φZ/2), which peaks at dF*; (ii) the *envelope* exp(−dF/ξtZ), which is monotonic and shifts the full-current peak earlier; and (iii) the *nodes*, which are envelope-immune and set the common period. η and ρs^eff have no envelope and peak at dF*; the raw current peaks earlier; the detrended current recovers dF*.

Table 1.1. v2.0 correction and value ledger.

Item	v1.2 status	v2.0 action	Level
Ic full peak	claimed at dF*	Theorem A14.8′: envelope-shifted, tan(AdF^pk/2ξZ)=AξtZ/ξZ	DERIVED
Locking F-A14.6	full-peak locking	Theorem A14.9′: detrended-peak + node/period locking	DERIVED
ξZ from Ic	raw peak	node/period or detrended prefactor (not raw peak)	DERIVED
Peak shift	unused	Theorem A14.11: ξtZ from peak shift (2nd channel)	DERIVED/TESTABLE
A14.7 status	DERIVED	DERIVED-CONDITIONAL ($\mathscr{A}_i$Z physical reality testable)	relabel
Ref [2]	“Cohen et al.”	“Yates et al.” (arXiv:1606.08619)	fix
BIC	co-headline	demoted to supporting; value on thickness-scan	reframe
Script header	Target 38/38	Target 47/47 (actual)	fix

§2. Inherited Corrections (C-01 … C-04)

The four v1.2 corrections are retained and re-verified. C-01: NEW-2 reads $\cos(\Theta SM/2) = \frac{1}{2}\,\mathrm{Tr}\, U\_AW[C]$ (half-angle, not its square), returning $\Theta SM = \mathbf{A}$. C-02: Lemma A14.B (clean single-cell interface) makes the universality $\Theta SM = \mathbf{A}$ precise via SU(2) conjugacy invariance of the trace. C-03: the SOC dephasing rate vanishes in the bulk ($\varepsilon \to 1$) and equals $2\Gamma so^0 \sin^2\beta$ near the core ($\varepsilon \to 0$); the v1.1 "bulk limit" label was wrong. C-04: the triplet-length ratio is quoted under the explicit convention $\gamma_0 = \Gamma so^0/(2\pi kBT + \Gamma sf) = 1$. All four pass unchanged (C-01 … C-04).

§3. Theorem A14.7 — Z-Spin SU(2)-Covariant Usadel Equation (DERIVED-CONDITIONAL)

External grounding (IMPORTED-PROVEN). Bergeret–Tokatly [7,8] showed that linear-in-momentum spin-orbit coupling enters the diffusive Usadel equation as a background SU(2) gauge field $\mathscr{A}_i = \mathscr{A}_i^a\, \sigma a/2$, and Tokatly [9] proved that the triplet relaxation is governed by $\Gamma^{ab} = D(\mathscr{A}\cdot\mathscr{A}\,\delta^{ab} - \mathscr{A}^a\mathscr{A}^b)$ (NEW-17).

Theorem A14.7 (DERIVED-CONDITIONAL). The Z-Spin sector contributes one additional, parameter-free SU(2) background field whose magnitude is fixed by the Wilson conjugacy angle **A** (A14.2′) and the Z-cell length ξZ,

$$\mathscr{A}_i Z = (\mathbf{A}/2\xi_Z)\, \hat{n}_Z\cdot\sigma\,, \quad \oint_{\mathrm{cell}} \mathscr{A}_i Z\, dl = (\mathbf{A}/2)\, \hat{n}_Z\cdot\sigma = -i \log U_Z[cell], \qquad \text{(NEW-18)}$$

$$D\, \tilde{\nabla}_i(\check{g}\, \tilde{\nabla}_i \check{g}) + [\, i\varepsilon\, \tau_3 + \check{\Delta} + \check{h}_Z\, ,\, \check{g}\,] = 0\,, \quad \tilde{\nabla}_i\, \check{g} = \partial_i\, \check{g} - i[\, \mathscr{A}_i so + \mathscr{A}_i Z\, ,\, \check{g}\,]. \qquad \text{(NEW-19)}$$

Verifications: (U-01) the one-cell holonomy of $\mathscr{A}_i Z$ reproduces the primitive Wilson loop NEW-1 to machine precision; (U-02) projecting Tokatly's Γ^{ab} onto the Z-aligned triplet axis gives $n\_a\, \Gamma^{ab}\, n\_b = D|\mathscr{A}so|^2 \sin^2\beta$, exactly the A14.4′ angular factor — a derived projection of an externally PROVEN tensor; (U-03) at $\beta = 0$ the covariant commutator $[\mathscr{A}so+\mathscr{A}Z, P\_Z]$ vanishes identically.

Status: DERIVED-CONDITIONAL. The equation *form* (NEW-19) is IMPORTED-PROVEN; the Z-Spin field $\mathscr{A}_i Z$ is a zero-parameter Z-Spin ansatz. What external theory proves is the SU(2)-covariant Usadel formalism and the SOC triplet-conversion structure, not the physical reality of $\mathscr{A}_i Z$; that reality is the testable content, checked through the locking and peak-shift predictions of §5–§6.1. **NC-A14.4:** A14.7 does not assert that $\mathscr{A}_i Z$ is an independently measured field; it asserts a parameter-free, falsifiable prediction for its magnitude.

§4. Theorem A14.8 — Z-Cell Current-Phase Relation, and A14.8′ — Envelope-Shifted Peak (DERIVED)

Theorem A14.8 (DERIVED). For a diffusive S/F/S junction the long-range triplet critical current is the A14.3 conversion prefactor times the A14.7 diffusive envelope (the damped-oscillatory form of Buzdin [10], IMPORTED-PROVEN):

$$I_c Z(d_F,\beta) = I_0 \sin^2(\mathbf{A}\, d_F/2\xi_Z) \cdot \exp[-d_F/\xi_t Z(d_F,\beta)], \quad \xi_t Z = \sqrt{(D/[2\pi k_B T+\Gamma_{sf}+\Gamma_{so} Z])}. \qquad \text{(NEW-20)}$$

§4.1 Theorem A14.8′ — Envelope-Shifted Peak

Theorem A14.8′ (DERIVED). Because the envelope is monotonically decreasing, the full current peaks *before* the conversion optimum. Setting d(ln IcZ)/ddF = 0 with ξtZ slowly varying gives the exact transcendental condition

$$(\mathbf{A}/\xi_Z) \cot(\mathbf{A}\, d_F peak/2\xi_Z) = 1/\xi_t Z \;\; \Leftrightarrow \;\; \tan(\mathbf{A}\, d_F peak/2\xi_Z) = \mathbf{A}\, \xi_t Z/\xi_Z. \qquad \text{(NEW-24)}$$

Two limits fix the interpretation. (i) Strong decay (ξtZ ≪ ξZ/**A**): dF^peak → 0, the current is envelope-dominated. (ii) Weak decay (ξtZ ≫ ξZ/**A**): tan → ∞, dF^peak → dF* = (π/**A**)ξZ, recovering the conversion optimum. Verifications: (D-01) the full peak is at 5.85 ξZ (β=0) and 3.06 ξZ (β=π/4), both < dF* = 39.23 ξZ; (D-02) NEW-24 holds at the numerical peak to 0.2%; (D-04) as ξtZ → ∞ the peak returns to dF*. The detrended current Ĩc = IcZ exp(+dF/ξtZ) removes the envelope and peaks exactly at dF* (D-03). Status: DERIVED.

§5. Theorem A14.9′ — Three-Observable Locking, Envelope-Immune Form (DERIVED)

Theorem A14.9′ (DERIVED). The conversion efficiency, the superfluid weight, and the detrended current are three functions of the single phase φZ(dF) = **A** dF/ξZ and share the peak dF* = (π/**A**)ξZ:

$$\eta_{S\to T} = \sin^2(\mathbf{A}\, d_F/2\xi_Z), \;\; \rho_s eff = \rho_s S[1+(20/11)\mathbf{A}\, \eta], \;\; \tilde{I}_c = I_0\, \eta. \qquad \text{(L)}$$

The locking is an exact algebraic identity (L-02), with α = dimZ·(Q−1)/Q = 20/11 inherited from ZS-F5 / ZS-A12:

$$\rho_s eff(d_F) - \rho_s S = (20/11)\, \mathbf{A}\, \rho_s S \cdot \eta_{S\to T}(d_F). \qquad \text{(NEW-22)}$$

Crucially, the raw current is locked through its *nodes*, not its peak: sin²(AdF/2ξZ) = 0 at dF = n·2πξZ/**A** regardless of the envelope, so the zeros of IcZ coincide with the η-zeros and the minima of ρs^eff (verification L-03; IcZ at the first node = 5×10^{-44}). The common period is therefore

$$\Delta d_F = 2\, d_F{}^* = 2\pi\xi_Z/\mathbf{A} \quad (\text{envelope-immune}). \qquad \text{(NEW-25)}$$

This is the corrected, robust form of the locking theorem: a single geometric constant **A** fixes the conversion angle, the conversion efficiency, the stiffness enhancement and the current period. Gate F-A14.6 is restated as node/period locking (Table 8.2), not full-peak locking. Status: DERIVED.

§6. Theorem A14.10 — Z-Sector Coherence-Length Extraction (DERIVED/TESTABLE)

Theorem A14.10 (DERIVED). ξZ is fixed by the envelope-immune node spacing of any locked observable, or equivalently by the prefactor/detrended peak of η, ρs^eff or Ĩc:

$$\xi_Z = \boldsymbol{A}\,\Delta d_F/2\pi \quad \textit{(node/period, all three observables)}, \quad \xi_Z = (\boldsymbol{A}/\pi)\, d_F{}^* \quad \textit{(peak of } \eta, \rho_s eff, \tilde{I}c\textit{)}. \qquad \text{(NEW-23)}$$

The raw-current peak must *not* be used for ξZ (it is envelope-shifted, A14.8′); the node method is preferred because it is envelope-immune and common to all three observables. Three independent extractions of ξZ (from η, ρs^eff, Ĩc) must agree; disagreement falsifies the locking theorem (Gate F-A14.6).

§6.1 Theorem A14.11 — Two-Channel Z-Length Spectroscopy

Theorem A14.11 (DERIVED/TESTABLE). The peak shift is not noise — it is a second observable. Inverting NEW-24 turns the measured full-current peak into the Z-aligned triplet length:

$$\xi_t Z = (\xi_Z/\boldsymbol{A})\, \tan(\, \boldsymbol{A}\ d_F peak/2\xi_Z\,). \qquad \text{(NEW-26)}$$

A single thickness scan therefore yields *two* Z-sector lengths: ξZ from the node spacing (NEW-23) and ξtZ from the peak shift (NEW-26). Verifications S-01 and S-02 recover both to 0.2% from the model scan. This upgrades A14 from a one-length probe to a Z-sector length spectrometer, and gives a parameter-free consistency check between the conversion channel (ξZ) and the triplet-transport channel (ξtZ). Status: DERIVED inversion; extracted values TESTABLE; first-principles ξZ remains O-A14.2 (OPEN).

§7. Diagnostic and Non-Claims

Axiom	Bridge	v2.0 closure
A3 Algebra (ℂ)	half-angle phase origin	A14.2′ + Lemma A14.B — CLOSED
A4 Non-triviality	topological triplet protection	A14.7 (DERIVED-CONDITIONAL) via Tokatly Γ^ab
A5 Unitarity	phase-stiffness ceiling	A14.9′ locking identity (envelope-immune)

NC-A14.5. The closed-form IcZ (A14.8) assumes the diffusive long-range-triplet regime [11,12,13]; ballistic, multidomain and strong-SOC corrections are out of scope. NC-A14.6. ξZ is not predicted numerically; A14.10/A14.11 extract it. NC-A14.7. “Peak at dF*” applies to the conversion prefactor and the detrended current only; the raw critical-current peak is envelope-shifted per A14.8′. NC-A14.8. The BIC analysis (Appendix B) is a posterior consistency check on existing spin-mixing-angle data (Yates et al. [2]), not a new-data prediction; the falsifiable value of A14 is carried by the thickness-scan (P-A14.7–P-A14.10).

§8. Experimental Predictions and Falsification Gates

Table 8.1. v2.0 predictions.

ID	Prediction	Gate
P-A14.7	Detrended current $\tilde{I}c=Ic\cdot\exp(+dF/\xi tZ)$ peaks at $dF^*=(\pi/A)\xi Z$	F-A14.8
P-A14.8	Nodes of Ic, η, ρs^eff coincide; common period $\Delta dF=2\pi\xi Z/A$	F-A14.6
P-A14.9	Three ξZ extractions (η, ρs^eff, $\tilde{I}c$) agree to ≤15%	F-A14.9
P-A14.10	Full Ic peak envelope-shifted: tan(AdF^pk/2ξZ)=AξtZ/ξZ; yields ξtZ	F-A14.10

Table 8.2. Falsification gates (v2.0).

Gate	Condition	Status
F-A14.6	Ic, η, ρs^eff nodes do not share period $\Delta dF=2\pi\xi Z/A$	OPEN — TESTABLE
F-A14.8	Detrended $\tilde{I}c(dF)$ peak deviates from dF* by >15%	OPEN — TESTABLE
F-A14.9	ξZ from the three observables disagree by >15%	OPEN — TESTABLE
F-A14.10	Full Ic peak does not satisfy tan(AdF^pk/2ξZ)=AξtZ/ξZ within 15%	OPEN — TESTABLE

§9. Verification Suite (47/47 PASS)

The 26 inherited tests (T-01…T-26) and four corrections (C-01…C-04) pass unchanged. v2.0 adds the Usadel block (U-01…U-03), the corrected current block (J-01…J-03), the new envelope-shift block (D-01…D-04), the corrected locking block (L-01…L-03), the ξZ block (X-01…X-02), and the peak-shift spectroscopy block (S-01…S-02). All use only the LOCKED inputs **A** = 35/437, **Q** = 11, dim(Z) = 2.

Table 9.1. v2.0 additions/changes to the verification suite (the 26+4 inherited tests pass unchanged).

ID	Statement	Result
U-01	Holonomy of 𝒜ᵢ^Z over one cell = UZ[cell]	PASS
U-02	Tokatly Γ^ab projected on n̂Z = sin²β (lifts A14.4′)	PASS
U-03	[𝒜so+𝒜Z, P_Z]=0 when SOC ∥ n̂Z	PASS
J-01	IcZ(0,β)=0 (no F layer)	PASS
J-02	Ic PREFACTOR peaks at dF*=π/A (not full current)	PASS

ID	Statement	Result
J-03	$I_c(\beta=0) \geq I_c(\beta=\pi/4)$ at fixed d_F	PASS
D-01	Full I_c peak envelope-shifted: $d_F^{pk} < d_F^*$ ($\beta=0, \pi/4$)	PASS
D-02	$\tan(A d_F^{pk}/2\xi_Z)=A\xi_{tZ}/\xi_Z$ at numerical peak ($\beta=0$)	PASS
D-03	Detrended $\tilde{I}_c$ peaks at $d_F^*=\pi/A$	PASS
D-04	$\xi_{tZ}\to\infty$ limit: full I_c peak $\to d_F^*$	PASS
L-01	η, ρ_s^{eff}, $\tilde{I}_c$ share peak at $d_F^*=\pi/A$	PASS
L-02	$\rho_s^{eff} - \rho_s^S = (20/11)A\cdot\eta$ exactly	PASS
L-03	Nodes of full I_c coincide with η nodes at $d_F=2\pi\xi_Z/A$	PASS
X-01	$\xi_Z=(A/\pi)d_F^*$ recovers ξ_Z (peak method)	PASS
X-02	$\xi_Z=A\cdot\Delta d_F/2\pi$ (node method, all three observables)	PASS
S-01	Peak-shift inversion recovers ξ_{tZ} from d_F^{pk}	PASS
S-02	Single scan yields both ξ_Z (nodes) and ξ_{tZ} (peak-shift)	PASS

Acknowledgements & Code Availability

This work was developed with the assistance of an AI collaborator (Anthropic Claude) for derivation-chain construction, numerical verification, external mathematical cross-reference, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. Verification script: zs_a14_v20_verify.py (header target and total both 47/47 PASS, exit code 0). All scripts at https://github.com/KennyKang-git/zspin.

Appendix A — New and Corrected Equations (v2.0)

Label	Equation	Status
NEW-17	$\Gamma^{ab} = D(\mathscr{A}\cdot\mathscr{A}\,\delta^{ab} - \mathscr{A}^a\mathscr{A}^b)$ (Tokatly 2017)	IMPORTED-PROVEN
NEW-18	$\mathscr{A}_i^Z = (A/2\xi_Z)\hat{n}_Z\cdot\sigma$; $\oint_{cell} \mathscr{A}^Z\,dl = (A/2)\hat{n}_Z\cdot\sigma$	DERIVED
NEW-19	Z-Spin SU(2)-covariant Usadel equation	DERIVED-CONDITIONAL
NEW-20	$I_{cZ} = I_0 \sin^2(A d_F/2\xi_Z)\, e^{(-d_F/\xi_{tZ})}$	DERIVED
NEW-22	$\rho_s^{eff} - \rho_s^S = (20/11)A\,\rho_s^S\,\eta$ (locking)	DERIVED
NEW-23	$\xi_Z = A\,\Delta d_F/2\pi = (A/\pi)d_F^*$	DERIVED
NEW-24	$\tan(A d_F^{pk}/2\xi_Z) = A\,\xi_{tZ}/\xi_Z$ (envelope-shifted peak)	DERIVED
NEW-25	$\Delta d_F = 2d_F^* = 2\pi\xi_Z/A$ (envelope-immune period)	DERIVED
NEW-26	$\xi_{tZ} = (\xi_Z/A)\tan(A d_F^{pk}/2\xi_Z)$ (peak-shift probe)	DERIVED

Appendix B — Supporting BIC Analysis (Yates et al. 2016)

The BIC comparison is retained as *supporting* evidence only (NC-A14.8). It uses five high-transparency spin-mixing-angle values from Yates et al. [2]; the dataset and the BIC definition are listed so that ΔBIC = 8.28 is reproducible from zs_a14_v20_verify.py (T-09). It is a posterior consistency check, not a new-data prediction.

Table B.1. Yates-2016 spin-mixing-angle dataset (rad); A = 0.080092; H_Z^(1): Θi = A (kZ=0).

i	Θi	σi	σint	Θi−A	χ² term
1	0.078	0.005	0.003	−0.00209	0.130
2	0.081	0.005	0.003	+0.00091	0.025
3	0.080	0.005	0.003	−0.00009	0.000
4	0.083	0.005	0.003	+0.00291	0.251
5	0.077	0.005	0.003	−0.00309	0.283

$$\Delta BIC = (\chi^2_{\mathrm{free}}+k_{\mathrm{free}}\ \ln N) - (\chi^2_Z+k_Z\ \ln N) = 8.28 > 2,\ \ \chi^2_Z=0.689,\ k_Z=0,\ N=5. \qquad \text{(B.1)}$$

Two hypotheses remain separated: H_Z^(1) (single-cell universality, Θi = **A**, kZ=0) — the DERIVED claim — and H_Z^(n) (multi-cell spectroscopy, Θi = ni**A**, ni∈ℕ discrete latent). A pre-registered Monte-Carlo control (seed 42, $N=5\times10^5$) gives a hit rate 0.64% < 5% for |θrand − **A**| < 0.010 rad.

References

Z-Spin Corpus

[ZS-F2] K. Kang, ZS-F2 v1.0: Geometric Impedance A = 35/437 (2026). [LOCKED]

[ZS-F4] K. Kang, ZS-F4 v1.0: Sector Contragredient Structure (2026). [V_XZ half-angle phase]

[ZS-F5] K. Kang, ZS-F5 v1.0: Gauge Symmetry Constraint Q = 11 (2026). [PROVEN: dim(Z)=2]

[ZS-M3] K. Kang, ZS-M3 v1.0: Regge-Holonomy (2026). [PROVEN: δφ = A, j = 1/2, ΠZ = sin²]

[ZS-S14] K. Kang, ZS-S14 v2.0: SM Master Action (2026). [SU(2)_L covariant derivative on H_5]

[ZS-A12] K. Kang, ZS-A12 v1.5: BEC/SC Unification (2026). [DERIVED-under-Regge; cardinal-2 ceiling; α=20/11]

[ZS-A14v1.2] K. Kang, ZS-A14 v1.2: Topic-Closure Release (June 2026). [Theorems A14.1–A14.10]

External Literature (APS / arXiv style)

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Version History

Version	Date	Changes
v1.0	June 2026	Initial release; Theorems A14.1–A14.5; 18/18 PASS.
v1.1	June 2026	Mathematical strengthening; A14.2′, A14.2-EXT, A14.3-EXT, A14.4′, A14.5′; BIC; 26/26 PASS.
v1.2	June 2026	Topic-closure; corrections C-01..C-04; Theorems A14.7–A14.10; 40/40 PASS.
v2.0	June 2026	Current-phase correction and value-maximizing release. (1) Theorem A14.8′ (envelope-shifted peak, tan(AdF^pk/2ξZ)=AξtZ/ξZ) corrects the v1.2 claim that the full Ic peaks at dF*. (2) Theorem A14.9′ restates locking in envelope-immune form (detrended-peak + node/period). (3) Theorem A14.11 (two-channel Z-length spectroscopy): single scan yields ξZ (nodes) and ξtZ (peak-shift). (4) A14.7 relabelled DERIVED-CONDITIONAL ($\mathscr{A}$Z physical reality testable). (5) Ref [2] corrected to Yates et al.; refs [12,13] added. (6) BIC demoted to supporting. (7) Equations NEW-24..NEW-26. Verification extended to 47/47 PASS; script header corrected. (Consolidated from internal Z-Spin Collaboration research notes up to v2.0.0.)